*Proof.* Suppose $f$ is continuous at $x$, and let $V$ be a neighborhood of $f(x)$. Since $V$ is open and contains $f(x)$, there is some $\varepsilon > 0$ with $B_\varepsilon(f(x)) \subseteq V$. By definition 2.5 there is some $\delta > 0$ with $f(B_\delta(x)) \subseteq B_\varepsilon(f(x)) \subseteq V$, so $U = B_\delta(x)$ is a neighborhood of $x$ with $U \subseteq f^{-1}(V)$.

Conversely, suppose the neighborhood condition holds, and let $\varepsilon > 0$. Then $V = B_\varepsilon(f(x))$ is a neighborhood of $f(x)$, so there is a neighborhood $U$ of $x$ with $U \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since $U$ is open and contains $x$, there is some $\delta > 0$ with $B_\delta(x) \subseteq U$, hence $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. □

**Theorem 2.2.** *A function $f : X \to Y$ between metric spaces is continuous iff $f^{-1}(V)$ is open in $X$ for every open $V \subseteq Y$.*

*Proof.* Suppose $f$ is continuous and $V \subseteq Y$ is open. For any $x \in f^{-1}(V)$, the set $V$ is a neighborhood of $f(x)$, so by theorem 2.1 there is a neighborhood $U$ of $x$ with $U \subseteq f^{-1}(V)$; thus every point of $f^{-1}(V)$ has a ball around it contained in $f^{-1}(V)$, which is therefore open.

Conversely, suppose preimages of open sets are open, and let $x \in X$. If $V$ is a neighborhood of $f(x)$, then $f^{-1}(V)$ is open by assumption and contains $x$, so it is itself a neighborhood of $x$ contained in $f^{-1}(V)$; by theorem 2.1, $f$ is continuous at $x$. □

Theorem 2.2 says that continuity between metric spaces can be detected using only their open sets, without any mention of distances. This motivates taking open sets themselves as the fundamental structure.

## 3 Topological spaces

**Definition 3.1** (topological spaces)**.** Given a set $X$, a *topology* on $X$ is a collection $\tau \subseteq 2^X$ of subsets of $X$ such that:

(1) the empty set and $X$ are open, i.e., $\emptyset, X \in \tau$;

(2) arbitrary unions of open sets are open, i.e., $\bigcup_\alpha U_\alpha \in \tau$ for any family $\{U_\alpha\}_\alpha \subseteq \tau$;

(3) the intersection of two open sets is open, i.e., $U \cap V \in \tau$ for any $U, V \in \tau$.

A set together with a topology $(X, \tau)$ is called a *topological space*. The elements of $\tau$ are its *open sets*.

**Proposition 3.1** (metric topology)**.** *The collection of open sets of a metric space $(X, d)$ is a topology on $X$, called the* metric topology *induced by $d$.*

*Proof.* The set $\emptyset$ is open vacuously, and $X$ is open since $B_1(x) \subseteq X$ for every $x \in X$. If $x \in \bigcup_\alpha U_\alpha$ for a family of open sets, then $x \in U_\alpha$ for some $\alpha$, so some ball around $x$ lies in $U_\alpha$ and hence in the union; arbitrary unions of open sets are therefore open. Finally, if $x \in U \cap V$ with $U, V$ open, then there are balls $B_\varepsilon(x) \subseteq U$ and $B_\delta(x) \subseteq V$, so $B_{\min(\varepsilon,\delta)}(x) \subseteq U \cap V$; the intersection of two open sets is therefore open. □

*Remark* 3.1. Not every topology arises from a metric; a space whose topology is induced by some metric is called *metrizable*.

**Example 3.1.** The *discrete topology* on a set $X$ is given by $2^X$, i.e., every subset of $X$ is open; it is induced by the discrete metric. The *trivial* (or *indiscrete*) *topology* has only $\emptyset$ and $X$ open.

**Corollary 3.1.** *The intersection of finitely many open sets is open.*

*Proof.* Induct on axiom (3) of definition 3.1: the two-set case is the axiom itself, and if $U_1, \dots, U_n$ are open, then $U_1 \cap \cdots \cap U_n = (U_1 \cap \cdots \cap U_{n-1}) \cap U_n$ is the intersection of two open sets. □